

EDUCATION / EXPERIENCE

Cornell University

Assistant professor in the School of Operations Research and Information Engineering

Ithaca, NY

2025–current

University of Pennsylvania

Postdoctoral researcher in the Department of Statistics and Data Science, Wharton School

Philadelphia, PA

2023–2025

Stanford University

Ph.D. in Statistics, Advisor: Andrea Montanari

Stanford, CA

2018–2023

Ph.D. Minor in Management Science and Engineering

2020–2023

M.S. in Statistics

2021–2022

Tsinghua University

B.S. in Mathematics

Beijing, China

2014–2018

RESEARCH INTERESTS

- Sampling in statistics and generative AI
- High-dimensional statistical inference
- Algorithm design under constraints
- Data-driven decision making
- Random matrix theory
- Mechanism design

JOURNAL PUBLICATIONS

- [1] Andrea Montanari* and Yuchen Wu* (2024). Provably efficient posterior sampling for sparse linear regression via measure decomposition. *Journal of the American Statistical Association* 2025.
- [2] Andrea Montanari* and Yuchen Wu* (2024). Fundamental limits of low-rank matrix estimation with diverging aspect ratios. *Annals of Statistics* 52, no. 4: 1460 - 1484.
- [3] Andrea Montanari* and Yuchen Wu* (2024). Statistically optimal first order algorithms: a proof via orthogonalization. *Information and Inference: A Journal of the IMA* 13, no. 4: iaae027.
- [4] Yuchen Wu* and Kangjie Zhou* (2024). Sharp analysis of power iteration for tensor PCA. *Journal of Machine Learning Research* 25, no. 195: 1–42.
- [5] Zihan Wei, Sarfaraz Alam, Miki Verma, Margaret Hilderbran, Yuchen Wu, Brandon Anderson, Daniel E Ho, Jenny Suckale (2024). Integrating water quality data with a Bayesian network model to improve spatial and temporal phosphorus attribution: Application to the Maumee River Basin. *Journal of Environmental Management* volume 360: 121120.
- [6] Andrea Montanari* and Yuchen Wu* (2023). Adversarial examples in random neural networks with general activations. *Mathematical Statistics and Learning* 6, no. 1: 143-200.
- [7] Song Mei* and Yuchen Wu* (2023). Deep networks as denoising algorithms: sample-efficient learning of diffusion models in high-dimensional graphical models. *arXiv:2309.11420. IEEE Transactions on Information Theory* 2025+.

CONFERENCE PUBLICATIONS

- [1] Yuchen Wu, Minshuo Chen, Zihao Li, Mengdi Wang, and Yuting Wei (2024). Theoretical insights for diffusion guidance: a case study for Gaussian mixture models. *International Conference on Machine Learning (ICML)*.
- [2] Pratik Patil*, Yuchen Wu*, and Ryan Tibshirani (2024). Failures and successes of cross-validation for early-stopped gradient descent. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, Oral, top 7.3% of accepted papers.
- [3] Yuchen Wu* and Kangjie Zhou* (2023). Lower bounds for the convergence of tensor power iteration on random overcomplete models. *Conference on Learning Theory (COLT)*.
- [4] Yuchen Wu, Jakab Tardos, Mohammad Hossein Bateni, André Linhares, Filipe Miguel Goncalves de Almeida, Andrea Montanari, Ashkan Norouzi-Fard (2021). Streaming belief propagation for community detection. *Advances in Neural Information Processing Systems (Neurips)*.
- [5] Michael Celentano*, Andrea Montanari* and Yuchen Wu* (2020). The estimation error of general first order methods. *Conference on Learning Theory (COLT)*.

PREPRINTS

- [1] Daniel Cao*, August Chen*, Karthik Sridharan*, Yuchen Wu* (2026). On the Robustness of Langevin Dynamics to Score Function Error. *arXiv:2603.11319*.
- [2] Sharan Sahu, Arisina Banerjee, Yuchen Wu (2026). Provably Reliable Classifier Guidance via Cross-Entropy Control. *arXiv:2601.21200*.
- [3] Yuchen Wu, Xinyi Zhong and Zhuoran Yang (2025). Learning to Lead: Incentivizing Strategic Agents in the Dark. *arXiv:2506.08438*.
- [4] Yuchen Wu, Yuxin Chen and Yuting Wei (2024). Stochastic Runge-Kutta methods: provable acceleration of diffusion models. *arXiv:2410.04760*.
- [5] Andrea Montanari* and Yuchen Wu* (2023). Posterior sampling in high dimension via diffusion processes. *arXiv:2304.11449*.

* alphabetical

SCHOLARSHIPS AND AWARDS

• ICSA China Conference Travel Award	2023
• SIAM Student Travel Award	2022
• National Scholarship, Tsinghua University	2015–2017
• Chinese Mathematical Olympiad, Second prize	2014
• Chinese Girls' Mathematical Olympiad, 3rd place	2013

TALKS AND PRESENTATIONS

1. ASA Section on Statistical Learning and Data Science, New York City	November 2026
2. INFORMS Annual Meeting	November 2026
3. ICSA Canada Symposium, McGill University	August 2026

4. Mathematical Foundations of Diffusion Model Workshop, IMSI	August 2026
5. IMS Annual Meeting, Salzburg, Austria	July 2026
6. Joint Conference on Statistics and Data Science, Guiyang, China	July 2026
7. 2026 ICSA China Conference, Shenzhen, China	June 2026
8. 2026 IMS-Asia Pacific Rim Meeting	June 2026
9. 2026 Applied Statistics Symposium, Arlington, Virginia	June 2026
10. Information Systems Laboratory Colloquium, Stanford University	May 2026
11. International Conference on Artificial Intelligence and Machine Learning, San Francisco	April 2026
12. Statistics Seminar, New Jersey Institute of Technology	April 2026
13. 2026 INFORMS Optimization Society Conference, Atlanta, Georgia	March 2026
14. Agent-AI Workshop by the Department of EECS, Syracuse University	February 2026
15. Colloquium on Statistics and Data Science, Tsinghua University	January 2026
16. Statistics Seminar, Central China Normal University	January 2026
17. Statistics and Data Science Seminar, Sanming University	January 2026
18. International Conference on Statistics and Data Science, Seville, Spain	December 2025
19. Statistics and Data Science Seminar, Beijing Normal University	December 2025
20. Statistics and Data Science Seminar, Peking University	December 2025
21. CAM Colloquium, Cornell University	November 2025
22. Probability Seminar, Cornell University	November 2025
23. Nxt AI-2025, Online	November 2025
24. INFORMS Annual Meeting, Atlanta, Georgia	October 2025
25. Statistics Graduate Society Seminar, Cornell University	September 2025
26. Statistics and Data Science Seminar, Cornell University	September 2025
27. INFORMS Applied Probability Society Conference, Georgia Tech	June 2025
28. International India Statistical Association Conference, University of Nebraska-Lincoln	June 2025
29. Center for Optimization and Statistical Learning Seminar, Northwestern University	May 2025
30. Statistics Meets Tensors Workshop, Institute for Mathematical and Statistical Innovation	May 2025
31. Optimization and Statistical Learning Workshop, Columbia University	April, 2025
32. Online MCMC Seminar	April, 2025
33. Applied Probability Seminar, Columbia University	October, 2024
34. FDS Conference: Recent Advances and Future Directions for Sampling, Yale University	October, 2024
35. INFORMS Annual Meeting	October, 2024
36. Cornell Young Researchers Workshop	October, 2024
37. S. S. Wilks Memorial Seminar in Statistics, Princeton University	October, 2024
38. Joint Statistical Meetings	August, 2024
39. The 41st International Conference on Machine Learning (poster)	July, 2024
40. Youth in High Dimensions workshop at ICTP	May, 2024
41. Professor Weijie Su's group meeting, University of Pennsylvania	May, 2024
42. The 27th International Conference on Artificial Intelligence and Statistics (oral)	May, 2024
43. The 58th Annual Conference on Information Sciences and Systems	March, 2024
44. Measure transport, diffusion processes and sampling (poster), Flatiron Institute	December, 2023
45. Professor Tom Berrett and Professor Yi Yu's group meeting, University of Warwick	November, 2023

46. IMS Young Mathematical Scientists Forum, University of Singapore	November, 2023
47. Wharton lunch seminar	November, 2023
48. Penn/Temple Probability Seminar	October, 2023
49. INFORMS Annual Meeting	October, 2023
50. University of the Chinese Academy of Sciences	October, 2023
51. Mathematical and Scientific Foundations of Deep Learning Annual Meeting (poster)	September, 2023
52. Theory lunch, Stanford University	August, 2023
53. University of Science and Technology of China	July, 2023
54. Zhongnan University of Economics and Law	July, 2023
55. Conference on Learning Theory 2023	July, 2023
56. ICSA 2023 China Conference	July, 2023
57. Shenzhen Conference on Random Matrix Theory and Applications	June, 2023
58. Yuxin Chen's group meeting	May, 2023
59. Ryan Tibshirani's group meeting	April, 2023
60. MoDL meeting	March, 2023
61. Liza Levina and Ji Zhu's group meeting, University of Michigan	January 2023
62. Institute for the Foundations of Data Science, Yale University	December 2022
63. Information Systems Laboratory Colloquium at Stanford University	December 2022
64. Stanford Berkeley Joint Colloquium	November 2022
65. SIAM Conference on Mathematics of Data Science	September 2022
66. TBSI Workshop on Learning Theory, Young Researchers' Forum session	August 2022
67. 2022 ICSA China Conference	July 2022
68. AI TIME PhD, Tsinghua University	February 2022
69. Yuling Jiao's group meeting, Wuhan University	January 2022
70. Conference on Neural Information Processing Systems	December 2021
71. No-retreat day student seminar, Department of Statistics, Stanford University	November 2021
72. 2021 Joint Statistical Meetings, speed presentation	August 2021
73. Conference on Learning Theory	July 2020

TEACHING

As an instructor at Cornell University:

- ORIE 7790 Selected Topics in Applied Statistics Spring 2026
- ORIE 6700 Statistical Principles Fall 2025

As an instructor at University of Pennsylvania:

- STAT 1010 Introductory Business Statistics Summer 2024

As a teaching assistant at Stanford University:

- STATS 200 - Statistical Inference Autumn 2018-2019, 2020-2021
- STATS 216 - Introduction to Statistical Learning Winter 2018-2019
- STATS 60 - Introduction to Statistical Methods Summer 2018-2019, 2019-2020, 2021-2022

• Math 230A / Stat 310A - Theory of Probability	Autumn 2019-2020
• STATS 218 - Introduction to Stochastic Processes II	Spring 2019-2020
• Math 230B / Stat 310B - Theory of Probability	Winter 2020-2021
• Math 230C / Stat 310C - Theory of Probability	Spring 2020-2021
• STATS 214 / CS 229M - Machine Learning Theory	Autumn 2021-2022
• STATS 217 - Introduction to Stochastic Processes I	Winter 2021-2022
• STATS 203 - Introduction to Regression Models and Analysis of Variance	Spring 2021-2022
• STATS 305B - Applied Statistics II	Winter 2022-2023

VISITING EXPERIENCE

- Visiting graduate student at Simons Institute
Program: Geometric Methods in Optimization and Sampling Fall 2021
- Visiting graduate student at the Institute for Advanced Study December 2022

PROFESSIONAL SERVICE

Grant reviewer:

Swiss National Science Foundation

Journal reviewer:

Annals of Statistics, Journal of the American Statistical Association, Biometrika, Annals of Applied Probability, Journal of Machine Learning Research, IEEE Transactions on Information Theory, Journal of Statistical Physics, SIAM Journal on Mathematics of Data Science, Communications on Pure and Applied Mathematics, IEEE Transactions on Big Data, Journal of Medical Internet Research, Probability Theory and Related Fields, Information and Inference: A Journal of the IMA

Conference reviewer:

Conference on Learning Theory (COLT), Conference on Neural Information Processing Systems (Neurips), International Conference on Machine Learning (ICML), International Conference on Learning Representations (ICLR), International Conference on Artificial Intelligence and Statistics (AISTATS), International Conference on Algorithmic Learning Theory (ALT), IEEE Symposium on Foundations of Computer Science (FOCS), IEEE International Symposium on Information Theory (ISIT), International Colloquium on Automata, Languages and Programming (ICALP), Conference on Artificial Intelligence (AAAI)

SKILLS

- Languages: Mandarin (native), English (advanced)
 - 112 in Toefl IBT test, November 2016
 - 165 (verbal) + 170 (quantity) + 4 in GRE test, October 2016
- Programming: Python, R, Matlab, C++